



SANIYA BANO

AI / ML | Data Science | Generative AI
Pre-Final Year B.Tech, Electronics Engineering | HBTU Kanpur
+91-9918157032 | 250006002@hbtu.ac.in | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Pre-final year Electronics Engineering undergraduate specializing in Generative AI, LLMs, RAG pipelines, and applied Machine Learning / Data Science. Skilled in building end-to-end AI and ML systems using LangChain, FAISS, Groq, Scikit-learn, and Pandas. Seeking internship opportunities in AI/ML, Data Science, and GenAI.

EDUCATION

Year	Degree/Certificate	Institute	Result
2025-2028	B.Tech Electronics Engineering	Harcourt Butler Technical University, Kanpur	Pursuing
2022-2025	Diploma in Computer Science & Engineering	GGP Bahraich	81% (Rank 1)

TECHNICAL PROJECTS

News Research Tool

Python, LangChain, FAISS, Groq (LLaMA-3.3-70B), HuggingFace Embeddings, Streamlit [\[GitHub\]](#)

- Designed and developed an AI-powered research assistant using Retrieval-Augmented Generation (RAG) architecture, implementing a complete pipeline: URL-based document loading, recursive text splitting/chunking, HuggingFace sentence embeddings, and FAISS vector store indexing for semantic search.
- Integrated Groq's LLaMA-3.3-70B large language model via RetrievalQAWithSourcesChain for low-latency inference, enabling users to ask natural-language questions across multiple news article URLs and receive answers with cited source links for transparency and factual grounding.
- Engineered a Streamlit-based user interface with secure API key management (python-dotenv) and robust error handling for edge cases including empty vector stores, missing environment variables, and invalid URLs, ensuring a production-ready, fault-tolerant application.

Student Performance Predictor

Python, Flask, Scikit-learn, XGBoost, CatBoost, Pandas, NumPy [\[GitHub\]](#)

- Architected an end-to-end, modular machine learning pipeline (data ingestion, data transformation, model training, and prediction) to predict students' mathematics scores from demographic, educational, and academic performance features, following industry-standard project structuring practices.
- Built a preprocessing pipeline separating numerical features (median imputation and standard scaling) from categorical features (imputation, one-hot encoding, and scaling), and persisted the fitted preprocessor and trained model as reusable serialized artifacts for consistent inference.
- Trained and benchmarked 8 regression algorithms (Linear Regression, Decision Tree, Random Forest, Gradient Boosting, KNN, XGBoost, CatBoost, AdaBoost) with grid-search hyperparameter tuning, selected the top performer by R^2 score against a defined quality threshold, and deployed it through a Flask web application with a form-based prediction interface.

CineScope — AI Movie Recommendation System

Python, FastAPI, Streamlit, TF-IDF, Cosine Similarity, TMDB API [\[GitHub\]](#)

- Built a full-stack, AI-powered movie recommendation web application using TF-IDF vectorization and cosine similarity for content-based filtering, integrated with live data from the TMDB API and deployed to production on Render.
- Developed an asynchronous FastAPI backend paired with a Streamlit frontend supporting Trending, Popular, Top Rated, and Upcoming movie browsing, full movie detail pages, genre-based fallback recommendations, search, runtime, rating, and overview display.

Exploratory Data Analysis — Zomato Dataset

Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter [\[GitHub\]](#)

- Conducted end-to-end exploratory data analysis on a real-world restaurant dataset, performing data cleaning, missing-value handling, and univariate and bivariate statistical analysis using Pandas and NumPy.
- Created data visualizations with Matplotlib and Seaborn to uncover actionable trends and correlations across restaurant ratings, average cost, location, and cuisine popularity, translating raw data into clear business insights.

House Price Prediction

Python, Scikit-learn, HistGradientBoostingRegressor, GridSearchCV, Pandas

- Designed an end-to-end regression pipeline (imputation, ordinal/one-hot encoding, standard scaling) and tuned a HistGradientBoostingRegressor via GridSearchCV with k-fold cross-validation, evaluating performance using RMSE and R^2 and visualizing feature importance with Matplotlib.

INTERNSHIP & TRAINING

Software Development Intern — 45-Day Industrial Training

- Hands-on training in full-stack web development: database connectivity, CRUD operations, and UI optimization.

TECHNICAL SKILLS

AI / GenAI: LangChain, RAG, FAISS, HuggingFace Embeddings, Groq, OpenAI, Prompt Engineering

ML / Data Science: Scikit-learn, Pandas, NumPy, XGBoost, CatBoost, Pipelines, GridSearchCV, Feature Engineering, Model Evaluation, EDA, Matplotlib/Seaborn

Languages & Web: Python, SQL, JavaScript • FastAPI, Flask, Streamlit

Databases & Tools: MongoDB, ChromaDB, SQL Server • Git, GitHub, Render, VS Code

LEADERSHIP

Web Development Head — Association of Electronics Engineering, HBTU Kanpur | Led team for technical projects & events.

ACHIEVEMENTS & CERTIFICATIONS

- Ranked 1st in Diploma in CSE — GGP Bahraich • All India Rank 33 in CUET-UG 2025 (HBTU Admission)